

fundamentals of ggplot

Ward, A; Parsons, R.

```
# install.packages("ggplot2")
library(ggplot2)
library(here)
```

here() starts at /Users/Abbie1/Documents/Repositories/BI0E 176 DataScience4EEB/Fundamentals of

```
# Read in the data and store as "ci_np" data object
ci_np <- read.csv("ci_np.csv")

# display the first 6 rows
head(ci_np)
```

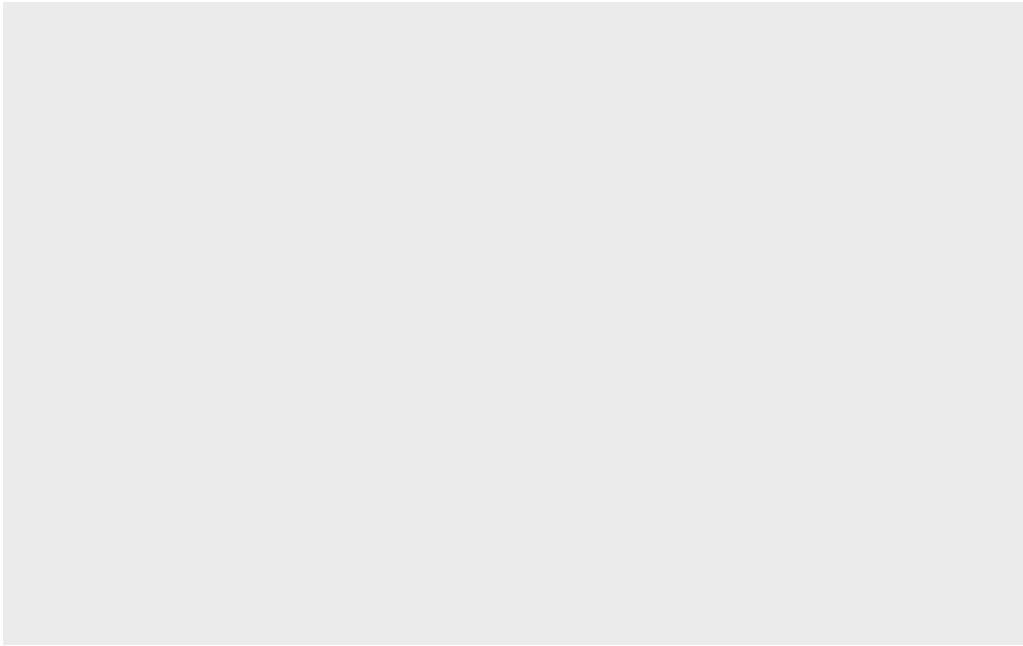
	region	state	code		park_name	type	visitors	year
1	PW	CA	CHIS	Channel Islands	National Park	National Park	1200	1963
2	PW	CA	CHIS	Channel Islands	National Park	National Park	1500	1964
3	PW	CA	CHIS	Channel Islands	National Park	National Park	1600	1965
4	PW	CA	CHIS	Channel Islands	National Park	National Park	300	1966
5	PW	CA	CHIS	Channel Islands	National Park	National Park	15700	1967
6	PW	CA	CHIS	Channel Islands	National Park	National Park	31000	1968

```
#look at column names
colnames(ci_np)
```

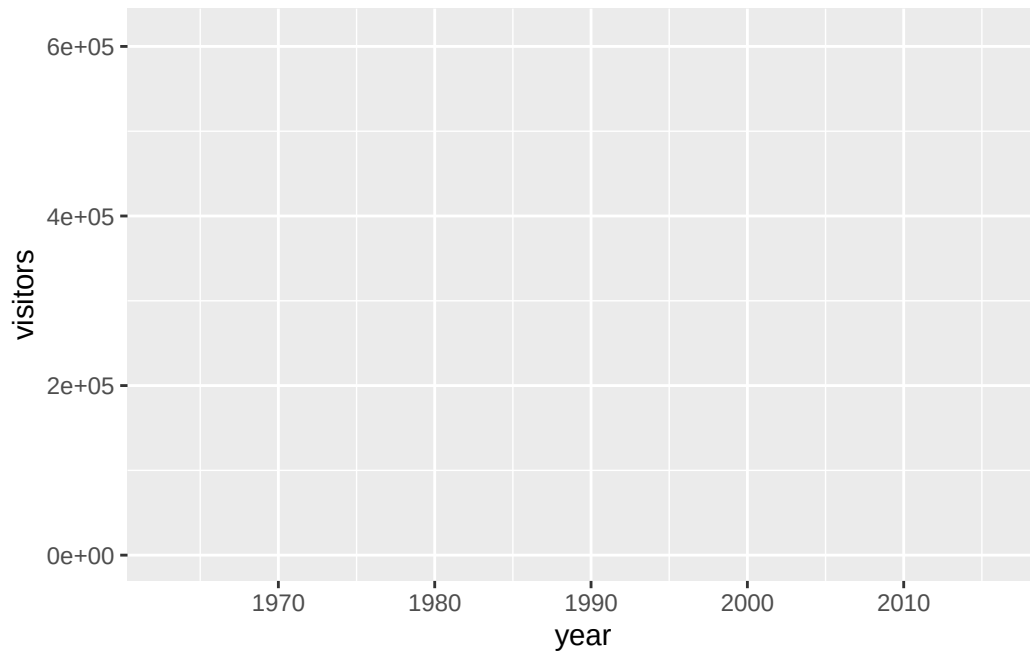
```
[1] "region"    "state"    "code"     "park_name" "type"     "visitors"
[7] "year"
```

Graphing Time YAY

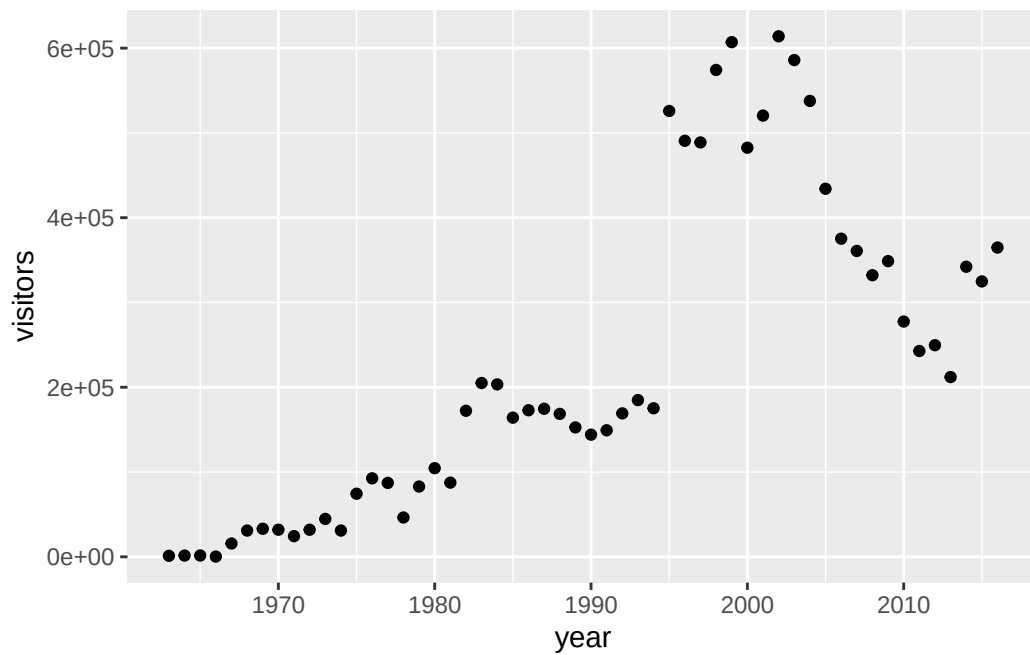
```
ggplot(data = ci_np)
```



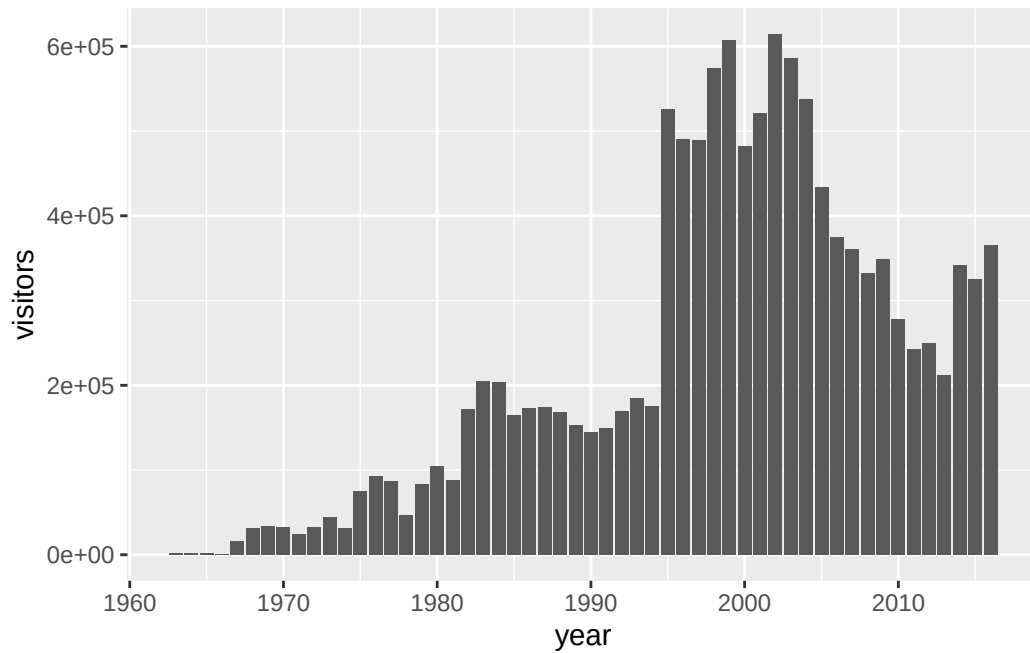
```
#try again w/ mapping  
ggplot(data = ci_np, mapping = aes(x = year, y = visitors))
```



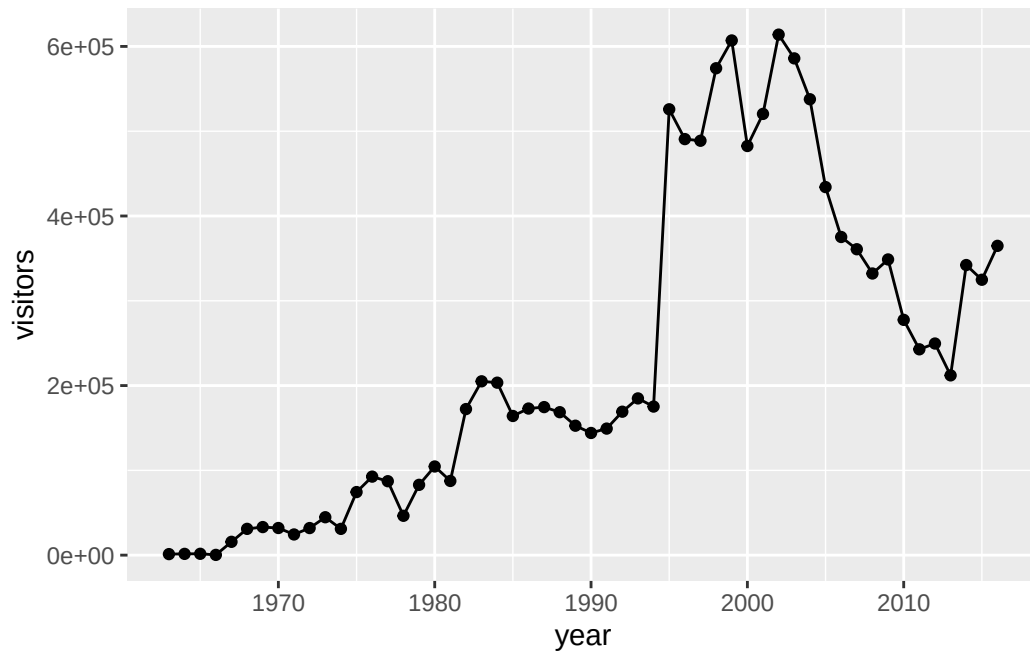
```
#Scatterplot  
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +  
  geom_point()
```



```
#Column graph
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_col()
```

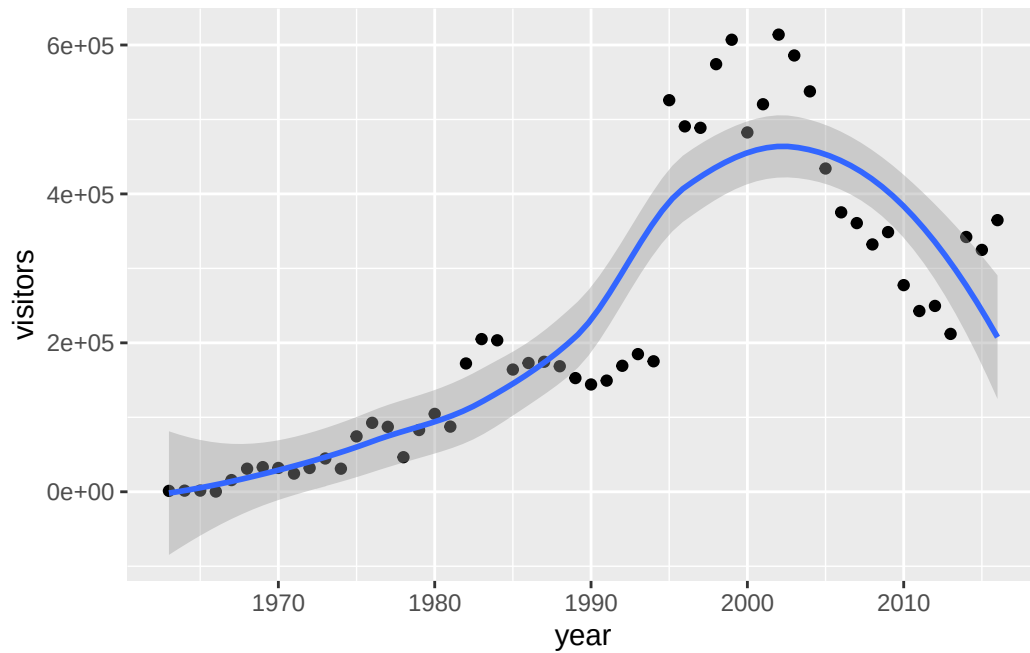


```
#Line plot
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_point() +
  geom_line()
```



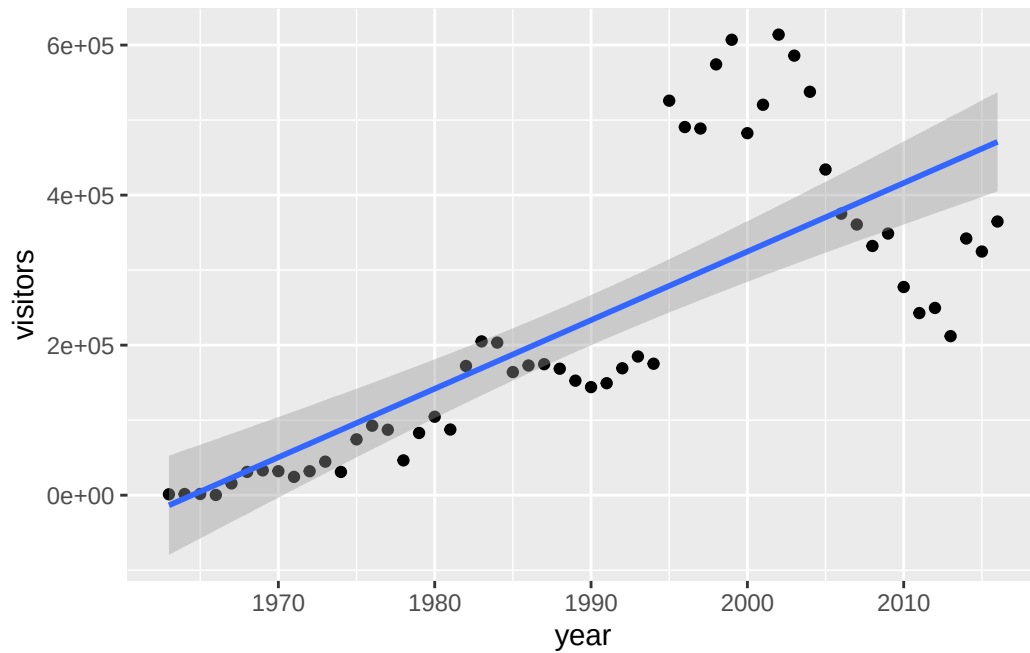
```
#Smooth line  
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +  
  geom_point() +  
  geom_smooth()
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



```
#Linear model
#Smooth line
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_point() +
  geom_smooth(method = "lm")
```

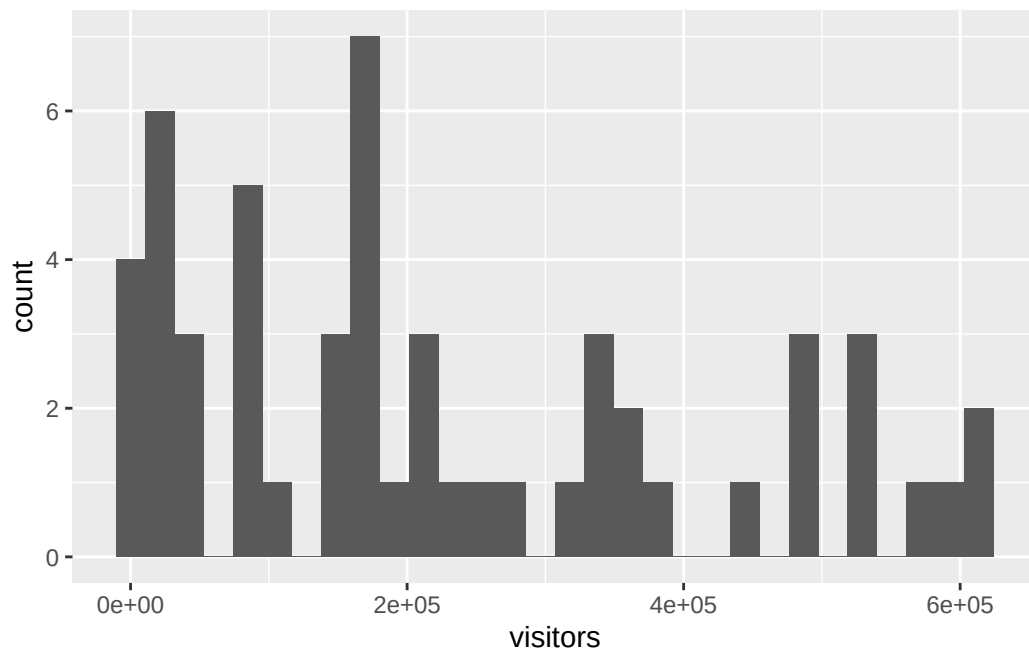
``geom_smooth()`` using formula = 'y ~ x'



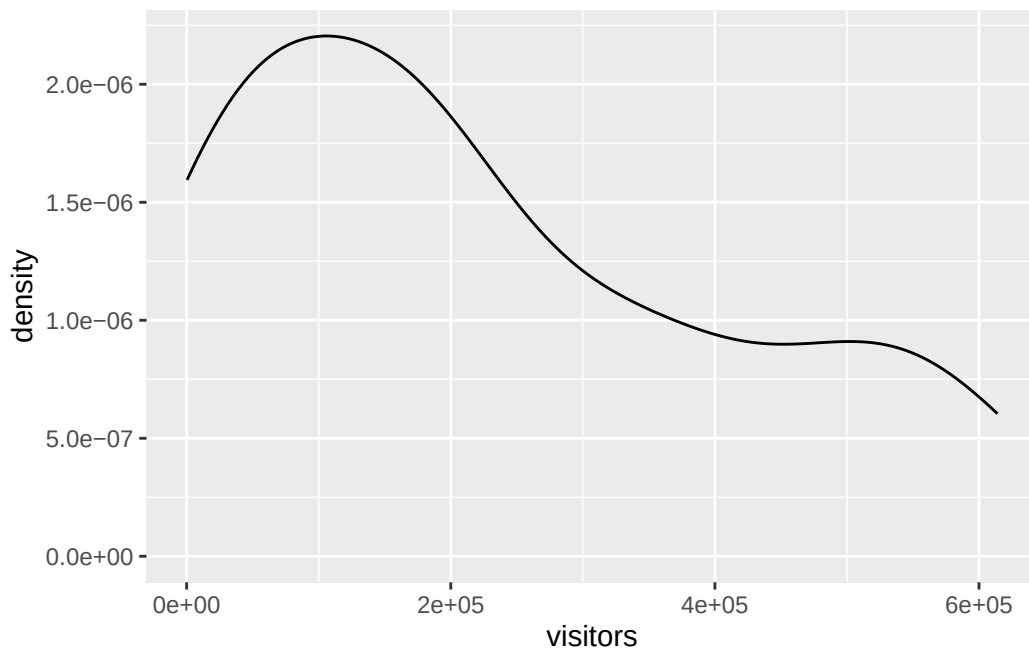
Single variables

```
#Histogram  
ggplot(data = ci_np, mapping = aes(x = visitors)) +  
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

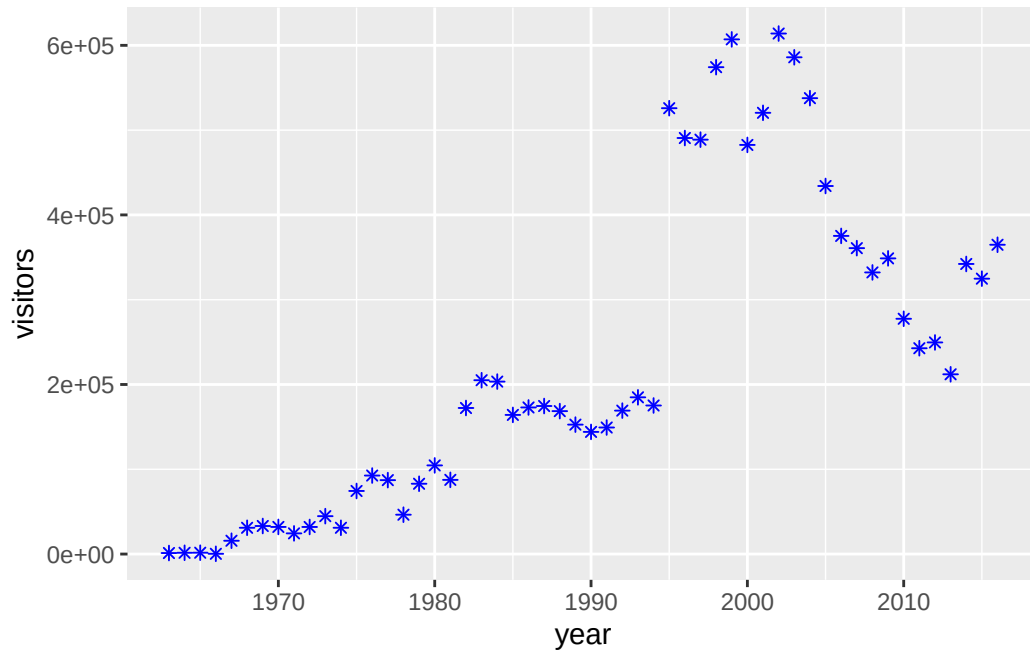


```
#Density Plot  
ggplot(data = ci_np, mapping = aes(x = visitors)) +  
  geom_density()
```

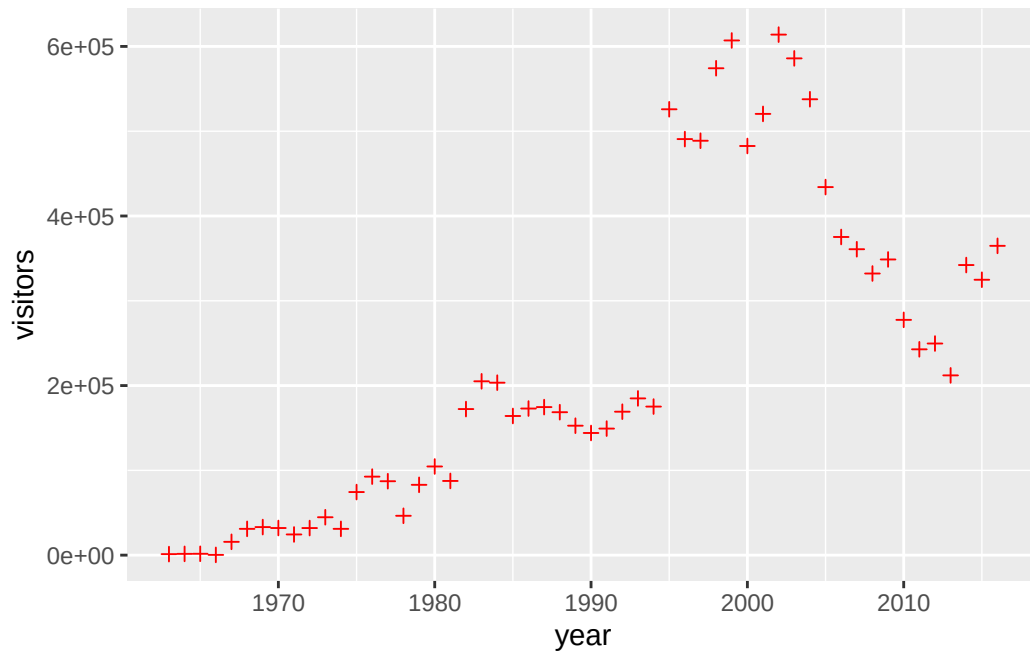


Modifying plot points

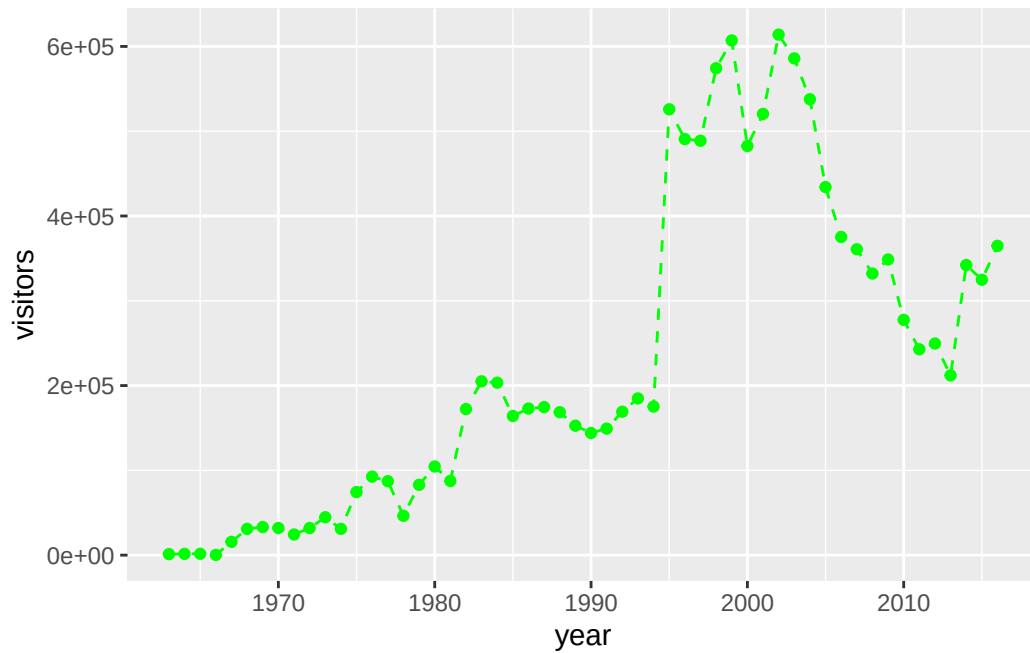
```
#Color and shape
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_point(shape = 8, color = "blue")
```



```
#Again!
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_point(shape = 3, color = "red")
```

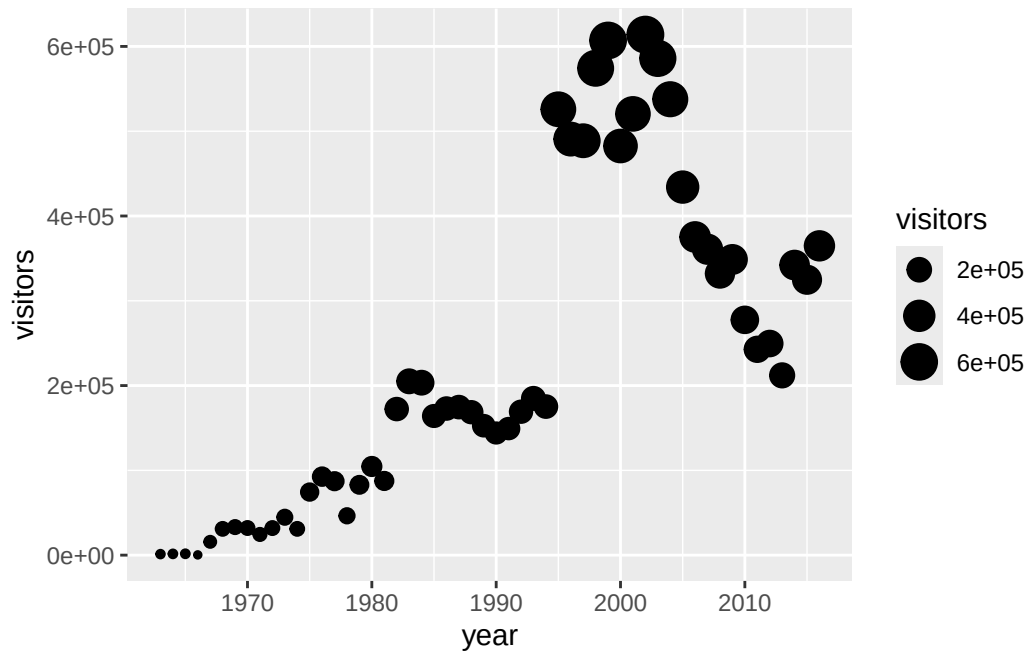


```
#Modifying line plot
ggplot(data = ci_np, mapping = aes(x = year, y = visitors)) +
  geom_point(color = "green") + #changed the dot color to be prettier
  geom_line(linetype = "dashed", color = "green")
```

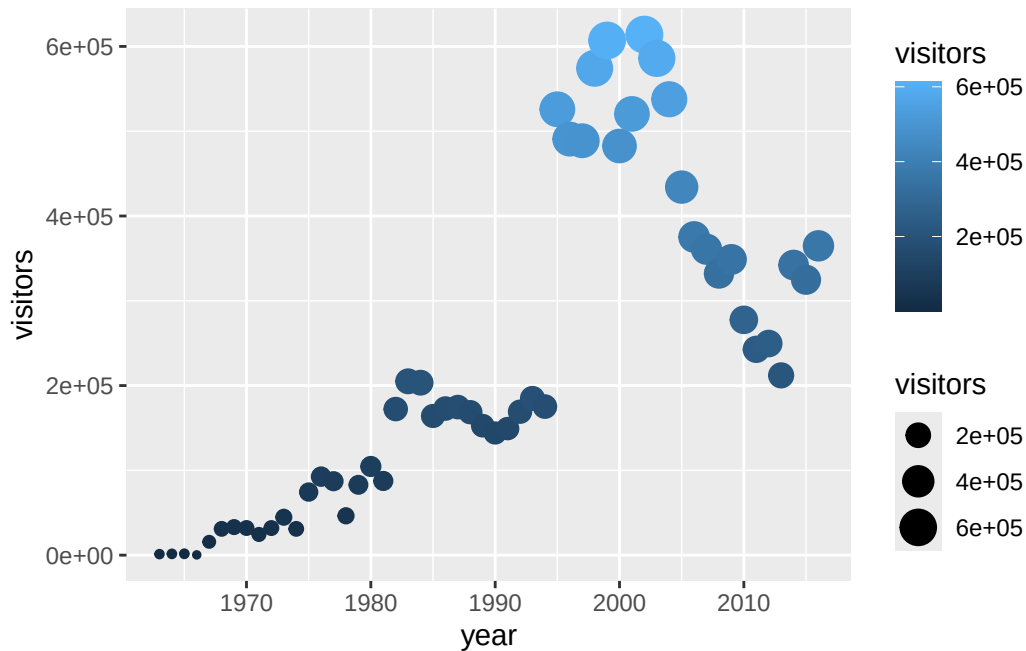


Mapping other variables onto aesthetics

```
#Size of points vary w/ visitors
ggplot(ci_np, aes(x = year, y = visitors)) +
  geom_point(aes(size = visitors))
```



```
#Changing color
ggplot(ci_np, aes(x = year, y = visitors)) +
  geom_point(aes(size = visitors,
                 color = visitors))
```



All CA!

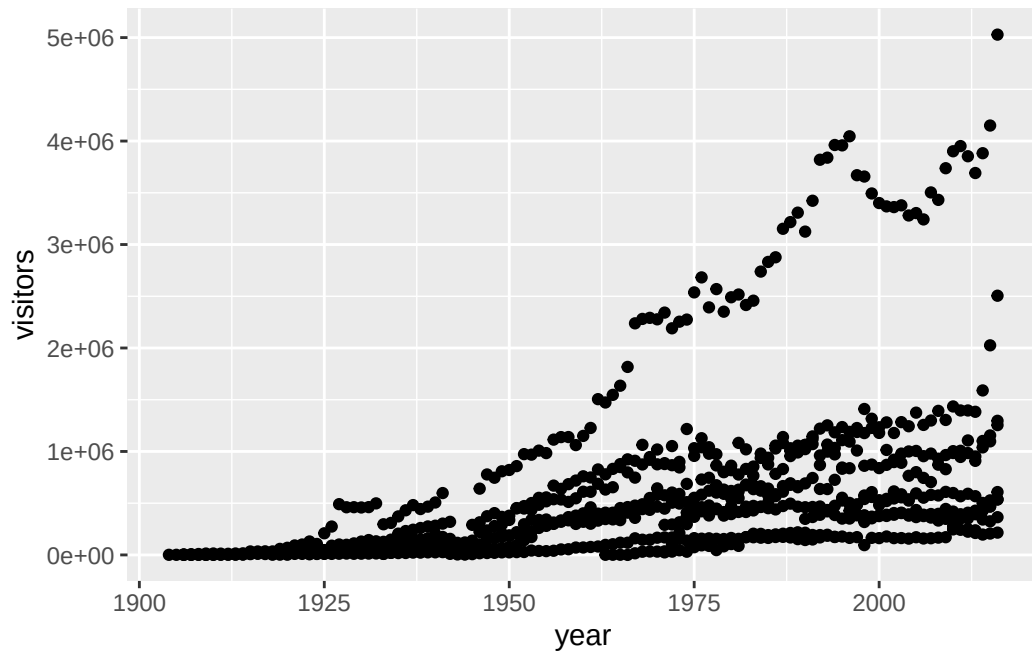
```
#Read in data
ca_np <- read.csv("ca_np.csv")

# Fetch the column but with the output run through the unique() function
unique(ca_np$park_name)
```

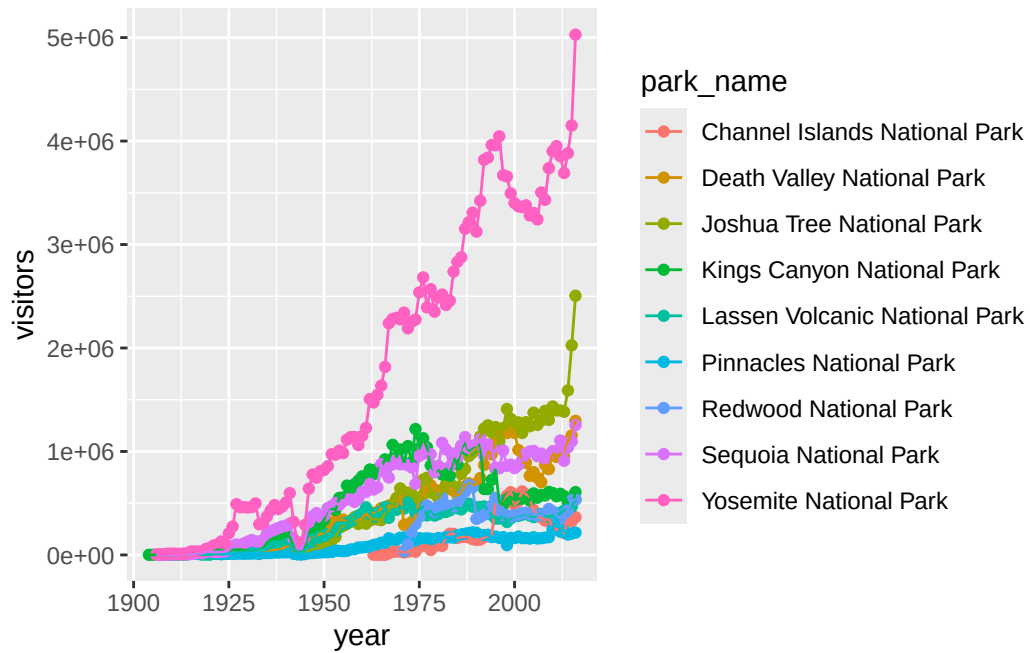
```
[1] "Channel Islands National Park" "Death Valley National Park"
[3] "Joshua Tree National Park"    "Kings Canyon National Park"
[5] "Lassen Volcanic National Park" "Pinnacles National Park"
[7] "Redwood National Park"       "Sequoia National Park"
[9] "Yosemite National Park"
```

Plots of visitors

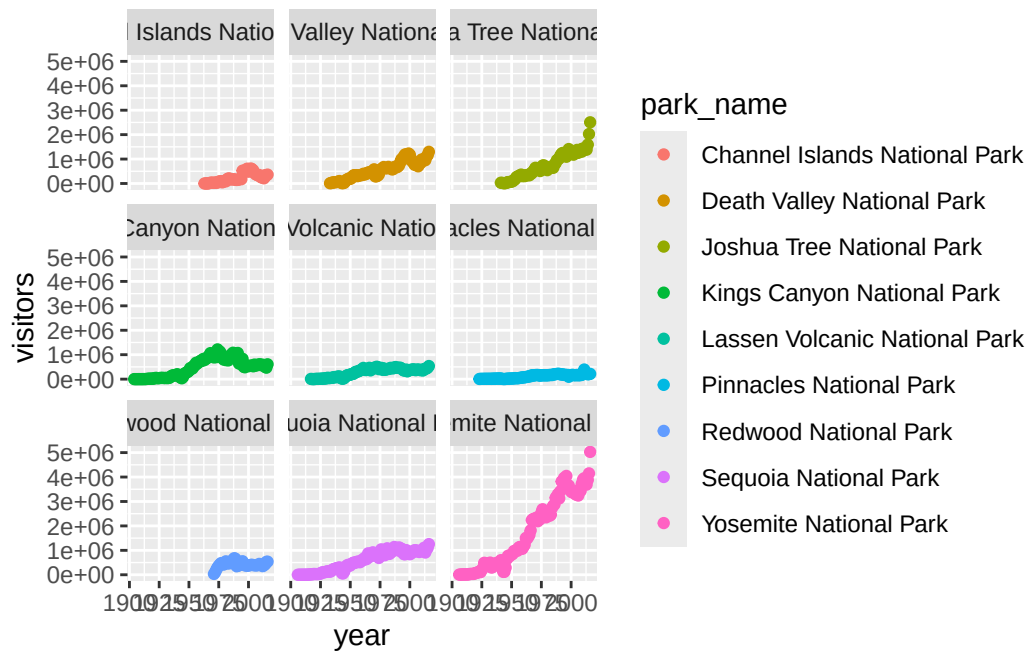
```
#visitors per park per year
ggplot(data = ca_np, mapping = aes(x = year, y = visitors)) +
  geom_point()
```



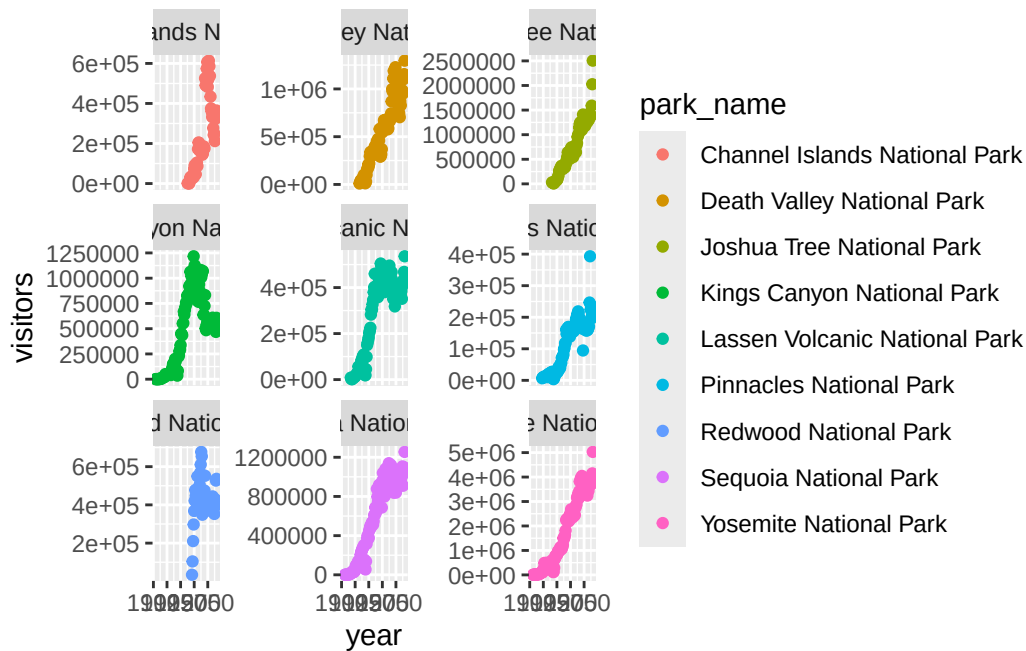
```
#color vary w/ park
ggplot(data = ca_np, mapping = aes(x = year, y = visitors, color = park_name)) +
  geom_point() +
  geom_line()
```



```
#Faceting parks
ggplot(data = ca_np, mapping = aes(x = year, y = visitors)) +
  geom_point(aes(color = park_name)) +
  facet_wrap(park_name ~ .)
```



```
#Free the scales
ggplot(data = ca_np, mapping = aes(x = year, y = visitors)) +
  geom_point(aes(color = park_name)) +
  facet_wrap(park_name ~ ., scales = "free_y")
```

Abalone Landings

```
# Read in the data
ab_land <- read.csv("abalone_landings.csv")

# display the first 6 rows
head(ab_land)
```

	Year	Abalone_Species	Abalone_Landings_lbs
1	1973	Black	1912519
2	1974	Black	1145396
3	1975	Black	684793
4	1976	Black	356951
5	1977	Black	463301
6	1978	Black	420045

```
#look at column names
colnames(ab_land)
```

```
[1] "Year" "Abalone_Species" "Abalone_Landings_lbs"
```

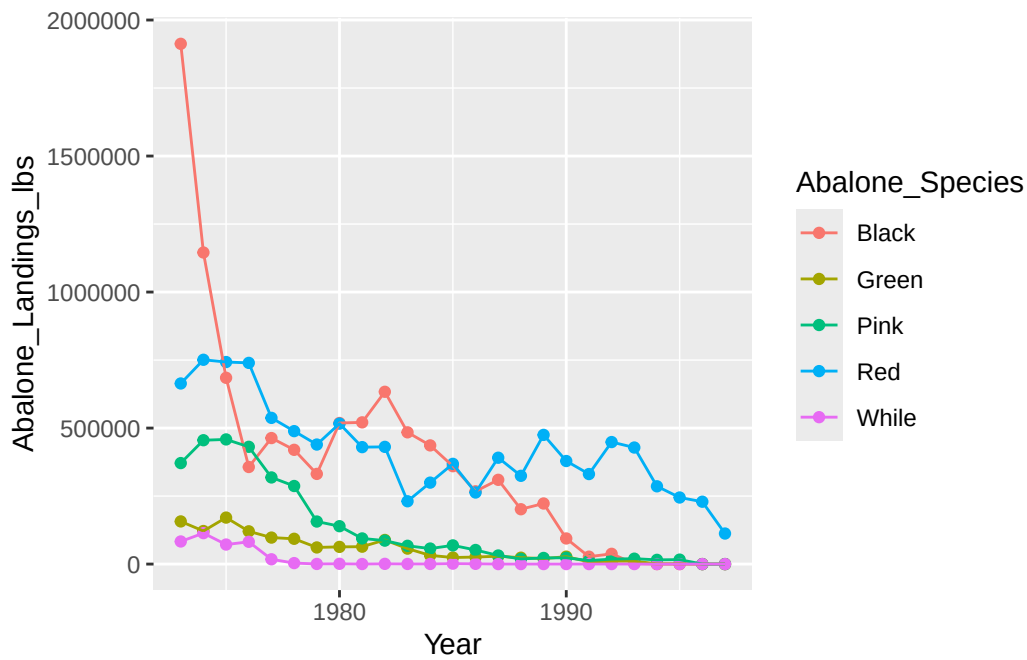
Data exploration notes:

- “While” should be “white”
- Landings by year per species easy comparison
- Changes in landings over time

Our research question:

How have landings for different species of abalone changed over time?

```
abalone_landings<-ggplot(ab_land, aes(x = Year, y = Abalone_Landings_lbs, color = Abalone_Sp  
  geom_point()+  
  geom_line()  
  
abalone_landings
```



```
ggsave("abalone_landings.jpeg", plot = abalone_landings)
```

Saving 5.5 x 3.5 in image